



Research Article

Clinical and EEG Analysis of Patients With Idiopathic Generalized Epilepsy and Photosensitivity

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Summary

Introduction: Our aim is to define the electroclinical characteristics of patients with photosensitive idiopathic generalized epilepsy (IGE) and to investigate the differences in seizure types, therapy and outcome between subgroups, based on photoparoxysmal response (PPR)/EEG characteristics and different syndromes.

Patients and Methods: Fifty consecutive patients with a diagnosis of IGE and having EEG or clinical evidence for photosensitivity have been identified from the epilepsy outpatient clinic database of the Department of Neurology (Istanbul Medical Faculty, Istanbul University). Clinical and EEG data was collected retrospectively.

Results: Thirty-six female and 14 male patients with median age 25.5 (14-63) were included to the study. The juvenile myoclonic epilepsy (JME) subgroup (n:24) had a significantly higher current age ($p=0.035$) and age at onset of seizures ($p=0.029$) compared with the absence epilepsy (AE) (n:12). Sleep deprivation triggers the seizures significantly more often in the JME subgroup compared with AE ($p=0.003$). Seizure records during EEG significantly more often in patients with AE ($p=0.016$). Patients with prolonged PPRs experienced significantly more clinical seizures during an EEG recording than patients with self-limited PPRs ($p=0.000$). The patients who were not seizure free had significantly more absence seizures (AS) (initially) compared with the patients who became seizure free ($p=0.001$).

Conclusion: Our findings in a retrospective cohort suggest that having AS is a negative prognostic factor for the seizure outcome in patients with photosensitive IGE.

Key words: PPR, photosensitivity, IGE

İdyopatik Fotosensitif Epilepsili Hastalarda Klinik ve EEG Bulguları

Özet

Giriş: Bu çalışmada amacımız fotosensitif idyopatik jeneralize epilepsili (İGE) hastaların elektroklinik özelliklerini tanımlamak ve nöbet tipleri, tedavi ve gruplar arasındaki ilişkileri fotoparoksizmal yanıt (FPY)/ EEG tipleri ve sendrom özelliklerine dayanarak belirlemektir.

Olgular ve Yöntem: Klinik olarak fotosensitif olan ya da EEG incelemelerinde fotosensitivite saptanan 50 İGE'li hasta İstanbul Üniversitesi, İstanbul Tıp Fakültesi, Nöroloji ABD, Epilepsi birimi polikliniği kayıtlarından retrospektif olarak saptanarak, klinik ve EEG bilgileri ayrıntılı olarak araştırıldı.

Sonuçlar: Ortalama yaşları 25.5 (14-63) olan 36 kadın, 14 erkek hasta çalışmaya dahil edildi. Juvenil miyoklonik epilepsili (JME) hastaların (n:24) mevcut yaşları ($p=0.035$) ve nöbet ortaya çıkma yaşları ($p=0.029$) absans epilepsili (AE) hastalarla (n:12) kıyaslandığında anlamlı ölçüde yüksekti. Uyku deprivasyonu; JME'li grupta nöbetleri AE'li gruba göre daha sık tetiklemekteydi ($p=0.003$). Stimülasyon sonrası süren uzun süreli FPY'ı olan hastalarda EEG çekimi sırasında klinik nöbet görülme oranı kısa süreli FPY görülenlere göre anlamlı

olarak yüksek bulundu ($p=0.000$). Ayrıca; çalışma sonunda nöbet geçirmeye devam etme olasılığı, nöbetsiz kalan grup ile karşılaştırıldığında başlangıçta absans nöbetleri olan hastalarda daha yüksekti ($p=0.001$).

Tartışma: Retrospektif bir kohort çalışması olan araştırmamız absans nöbetlerinin fotosensitif IGE'li hastalarda negatif bir prognostik faktör olduğunu düşündürmektedir.

Anahtar Kelimeler: FPY, fotosensitivite, İJE

INTRODUCTION

Photosensitivity (PS) is an abnormal sensitivity of the brain to visual/light stimuli. It can be detected with electroencephalogram (EEG) as a paroxysmal epileptiform reaction to intermittent photic stimulation (IPS), which was called as photoparoxysmal response (PPR)^(3,19,20) and have been demonstrated in epileptic as well as non-epileptic persons^(10,15,18,21). Studies in epileptic patients show that an epileptiform response to IPS is found in about 10-20% of children and 5-10% of adults. This response is more common in females at any age^(12,23).

It has been known that flickering sunlight can provoke epileptic seizures. Other typical trigger factors include TV, computer and video games, venetian blinds, striped walls, rolling stairs, striped clothing and natural and environmental lighting. Additional human factors such as cognitive abilities, emotional factors, fatigue and sleep deprivation may facilitate the occurrence of PPRs and seizures^(3,5,11,20). Individual factors such as age and gender and many other, unknown possibly genetic factors seem to influence the expression of the PPR⁽¹⁷⁾. The precise mechanisms of inheritance of PS are not known yet and seem to be complex, involving several genes, which probably interact with each other.

PS can be assessed with different diagnostic procedures, but the most common method is IPS⁽¹⁶⁾. Different protocols and photic stimulators are used across laboratories^(10,14). A widely used EEG classification system was proposed

by Waltz⁽²²⁾; subclassifying the PPR in four phenotypically different types:

PPR type I: spikes within the occipital rhythm

PPR type II: parieto-occipital spikes and biphasic slow waves

PPR type III: parieto-occipital spikes and biphasic slow waves spreading to frontal regions

PPR type IV: generalized spikes or polyspikes and waves

Prolonged (self-sustained) PPRs, which outlast the stimulus train, can be distinguished from self-limited PPRs, which cease spontaneously or when the flashes stop. Some studies suggest an association between prolonged PPRs and a higher incidence of seizures⁽¹⁴⁾. The relation between the type of IPS response and risk of visually induced seizures is still unclear⁽²⁰⁾.

The clinical signs and symptoms when exposed to photic stimuli can vary from mild subjective symptoms to all kinds of epileptic seizures like myoclonic seizures (MS), absence seizures (AS), generalized tonic-clonic seizures (GTCS) and focal seizures^(3,6,11). PS is most commonly associated with idiopathic generalized epilepsies (IGEs), such as juvenile myoclonic epilepsy (JME) and childhood absence epilepsy (CAE).

The aims of this study are to define the electroclinical characteristics of patients with photosensitive IGE and to study differences in seizure types, therapy and outcome between subgroups, based on PPR/EEG characteristics and different syndromes of idiopathic generalized epilepsy such as JME and AE.

MATERIAL AND METHODS

Patient selection

Fifty patients with a diagnosis of IGE and having EEG or clinical evidence for photosensitivity have been identified from the epilepsy outpatient clinic database of the Department of Neurology (Istanbul Medical Faculty, Istanbul University). EEG evidence for photosensitivity was obtained from available EEG traces.

Inclusion criteria: Patients with a diagnosis of IGE and having EEG or clinical evidence for PS, follow-up period ≥ 1 year (last follow-up in 2007), no evidence of neurologic deficit or mental retardation, CT/MR scans normal or nonspecific/unrelated to epilepsy if performed.

Study protocol

EEG protocol

IPS protocol: The standard IPS protocol starts with eyes open for 5 seconds and then eyes closed for 5 seconds; the stimulus trains of 5-10-15-20-25-30-35 Hz are given at a distance of 30 cm in a normally illuminated room. If PS is found or suspected, the highest and lowest Hz ranges are determined.

Defining and grading PS: A clear distinction is being made between interictal epileptiform EEG-discharges to IPS and clinical seizures provoked by IPS. In the situation where epileptiform discharges are already present during rest and hyperventilation (HV), PS is assumed if twofold or more increase during IPS. The IPS response is graded according to Waltz's classification⁽²²⁾ by the investigators.

Clinical data: The following clinical parameters were assessed from the outpatient medical records: diagnosis/syndrome (JME, AE or other IGE), sex, current age, age at onset of

epilepsy, follow-up period, past history/associated medical conditions, family history of epilepsy, febrile convulsions ever, status epilepticus ever, subjective symptoms ever, triggers in daily life, seizure types, therapy (AED) and seizure outcome. Remission was defined as seizure free for ≥ 5 years, regardless of AED treatment status.

EEG data: The following information was obtained from the available EEG reports/traces: age at first and last EEG, total number of EEGs, PPR type, % EEGs with a PPR, presence of interictal spontaneous epileptiform discharges, epileptiform discharges during HV, eye closure sensitivity, background activity, ictal findings, presence of a prolonged PPR and the range of IPS frequencies inducing a PPR.

Statistical analysis

The statistical analysis was done with SPSS for Windows version 15.0. For continuous variables, associations were evaluated using the Mann-Whitney U test for two independent samples. Associations between categorical variables were assessed by using the Chi-square (Pearson Chi-square) test. Analysis of the same patient group on two different dates was done by using the nonparametric McNemar test for two related samples. Statistical significance was set at $p < 0.05$.

RESULTS

The 50 patients with photosensitive IGE included 36 female and 14 male patients (female/male ratio = 2.6:1). Patient characteristics are shown in Table 1. The trigger factors are also listed in Table 1. The occurrence of sleep deprivation as a seizure trigger was reported significantly more often in the JME subgroup compared with the AE subgroup ($p = 0.003$).

Table 1:. Patient characteristics and seizure triggers of 50 patients with photosensitive IGE (at first consultation).

PATIENT CHARACTERISTICS	Patients				P-value
	JME N=24	AE N=12	IGE (other) N=14	Total N=50	
Median age years (min- max)	28.5 (20-63)	22.5 (20-60)	23 (14-44)	25.5 (14-63)	0.035
Median age at onset (min-max)	14 (6-25)	11.5 (2-23)	14.5 (1-18)	14 (1-25)	0.029
Median age at first consult (min-max)	18 (13-59)	17 (7-46)	17.5 (9-40)	17.5 (7-59)	NS
Median age at last consult (min-max)	27.5 (19-63)	22.5 (19-60)	22.5 (14-43)	25 (14-63)	NS
Female sex (%)	17/24 (71)	9/12 (75)	10/14 (71)	36/50 (72)	NS
head trauma (& loss of consciousness) (%)	6/24 (25)	1/12 (8)	0/14 (0)	7/50 (14)	NS
Psychiatric disorders (%)	5/24 (21)	3/12 (25)	2/14 (14)	10/50 (20)	NS
Family history of epilepsy (%)	11/24 (46)	2/12 (17)	10/14 (71)	23/50 (46)	NS
Status epilepticus ever (%)	0/24 (0)	0/12 (0)	2/14 (14)	2/50 (4)	-
Febrile convulsions ever (%)	3/24 (13)	2/12 (17)	1/14 (7)	6/50 (12)	NS
Subjective symptoms ever (%)	10/24 (42)	8/12 (67)	10/13 (77)	28/50 (56)	NS
sleep deprivation (%)	20/24 (83)	4/12 (33)	7/14 (50)	31/50 (62)	0.003
emotional factors/stress (%)	11/24 (46)	6/12 (50)	4/14 (29)	21/50 (42)	NS
TV (%)	11/24 (46)	5/12 (42)	11/14 (79)	27/50 (54)	NS
sun-light (%)	9/24 (38)	8/12 (67)	9/14 (64)	26/50 (52)	NS
Other	1/24 (4)	0/12 (0)	0/14 (0)	1/50 (2)	NS

AE: absence epilepsy, IGE: idiopathic generalized epilepsy, JME: juvenil myoclonic epilepsy, Max: maximum, Min: minimum, NS: not significant, PC: computer, TV: television, VG: video-game.

*less otherwise specified

At the first visit presentation, 46 (92%) patients had experienced one or more generalized tonic-clonic seizures (GTCS), at last follow-up only 4 (8%) patients reported still having GTCS ($p=0.000$). Valproic acid (VPA) constituted the AED therapy in the majority of the (62%) patients. At last follow-up, significantly less patients reported using other AED compared with the first consultation ($p=0.027$). Significantly more patients reported being medication free at last follow-up than at first consultation ($p=0.021$). No significant differences in

treatment status were found between the JME and the AE patients.

EEG data

The EEG findings are summarized in Table 2. The majority of the patients (42/50; 84%) expressed a PPR type IV during IPS. In 18 (36%) patients, ictal clinical phenomena occurred during an EEG registration and these were significantly more in patients of the AE subgroup, compared to the JME subgroup (67% compared to 25%; $p=0.016$). Most patients [38(93%)] were sensitive at a IPS-range of 10-20 Hz and 33 (81%) patients were sensitive at 20-30 Hz respectively.

Table 2: EEG findings in 50 patients with photosensitive IGE.

Patients	JME N=24	AE N=12	IGE (other) N=14*	Total N=50	P-value JME-AE
Median age at first EEG (min-max)	16 (9-59)	14 (7-45)	15 (1-40)	15 (1-59)	NS
Median age at last EEG (min-max)	23.5 (18-59)	20.5 (13-60)	22 (13-40)	22 (13-60)	NS
Median number of EEGs (min-max)	4 (1-10)	7.5 (1-14)	3 (1-16)	4 (1-16)	NS
Interictal spontaneous epileptiform discharges (%)	17/24 (71)	10/12 (83)	8/14 (57)	35/50 (70)	NS
Epileptiform discharges during HV (%)	9/24 (38)	6/12 (50)	4/14 (29)	19/50 (38)	NS
Normal background activity (%)	24/24 (100)	11/12 (92)	14/14 (100)	49/50 (98)	NS
ECS (%)	4/19 (21)	4/10 (40)	1/11 (9)	9/40 (23)	NS
Ictal clinical phenomena (%)	6/24 (25)	8/12 (67)	4/14 (29)	18/50 (36)	0.016
PPR type I (%)	1/24 (4)	0/12 (0)	0/14 (0)	1/50 (2)	NS
PPR type II (%)	0/24 (0)	1/12 (8)	2/14 (14)	3/50 (6)	NS
PPR type III (%)	3/24 (13)	0/12 (0)	1/14 (7)	4/50 (8)	NS
PPR type IV (%)	20/24 (83)	11/12 (92)	11/14 (79)	42/50 (84)	NS
Prolonged PPR (%)	11/23 (48)	6/10 (60)	5/13 (39)	22/46 (48)	NS
IPS-sensitivity range (Hz) 1-10 (%)	8/21 (38)	1/9 (11)	5/11 (46)	14/41 (34)	NS
IPS-sensitivity range (Hz) 10-20 (%)	18/21 (86)	9/9 (100)	11/11 (100)	38/41 (93)	NS
IPS-sensitivity range (Hz) 20-30 (%)	16/21 (76)	8/9 (89)	9/11 (82)	33/41 (81)	NS
IPS-sensitivity range (Hz) >30	3/21 (14)	2/9 (22)	4/11 (36)	9/41 (22)	NS

AE: absence epilepsy, ECS: eye closure sensitivity, HV: hyperventilation, IGE: idiopathic generalized epilepsy,

IPS: intermittent photic stimulation, JME: juvenil myoclonic epilepsy, Max: maximum, Min: minimum

NS: not significant, PPR: photoparoxysmal response. *less otherwise specified

Prolonged PPR versus self-limited PPR

22 (48%) of 46 patients showed prolonged PPRs and 24 (52%) self-limited PPRs on EEG. There were no significant differences between these 2 subgroups in any of the clinical, EEG and seizure characteristics. However, patients with prolonged PPRs experienced significantly more clinical seizures during an EEG recording than patients with self-limited PPRs ($p=0.000$).

Factors associated with seizure outcome

Significantly more AS (initially) were noted in the group of patients who were not seizure free compared to the group of patients who became seizure free ($p=0.001$). In addition, the combination of AS+MS+GTCS (initially) occurred significantly more in this patient group (0.009) and the combination of MS+GTCS (initially) occurred significantly more in the patient group that became seizure free ($p=0.003$).

Follow-up data

The median follow-up period was 7 years (range, 1-22 years) for the whole photosensitive IGE group, 9 years (range, 2-22 years) for the JME subgroup and 6.5 years (range, 2-14 years) for the AE subgroup. At last follow-up, 19 (38%) patients reported being seizure free (9 JME patients, 5 AE patients and 5 other IGE patients). Eight patients (16%) had achieved complete remission of seizures $\geq$ 5 years and 2 of them were no longer taking AEDs. There were no significant differences in seizure outcome/remission between the JME patients and AE patients. No relation was noted between seizure outcome and family history of epilepsy, past history, visual triggers, sex, current age, age at onset of seizures, ictal findings during EEG, interictal spontaneous epileptiform discharges and prolonged PPR.

DISCUSSION

Electroclinical characteristics of the patients with photosensitive IGE were investigated in this study. There is an

obvious female preponderance (2.6:1), in concordance with literature^(7,9).

All patients showed a PPR during IPS; the majority (42/50; 84%) expressed a PPR type IV. At least 80% of patients with photosensitive epilepsy have a history of seizures precipitated by environmental visual stimuli⁽²²⁾. Similar results were found in this group: 35 (70%) patients reported having seizures triggered by visual stimuli; especially TV and (sun)light.

There was a significant decrease in prevalence of all seizure types and 19 (38%) patients reported being seizure free at last follow-up. Eight patients (16%) had achieved complete remission of seizures $\geq$ 5 years and 2 of them were no longer taking AEDs. This might be explained by suppression of seizures by appropriate treatment or a possible favourable prognosis of photosensitive IGE. In literature, the prognosis of photosensitive epilepsy is not clear^(8,19). Harding et al.⁽⁸⁾ state that there is no particular age at which photosensitivity disappears and their study indicates a 14-37% probability that photosensitivity will disappear spontaneously. VPA is the drug of first choice in treating photosensitive patients^(8,9) and accordingly, in this cohort, the majority of patients used VPA to control their seizures.

Determination of IPS sensitivity range can be used as a measure of epileptogenesis. This range is related to liability of seizures in daily life, i.e. patients with a greater range have a higher likelihood of visually-induced seizures than those with a smaller range. In this cohort, most patients were sensitive at a IPS-range between 10-20 Hz and 20-30 Hz; 38 (93%) and 33 (81%) of 41 patients, respectively. This is in concordance with what is known in literature: the lower and upper limits are usually between 10 and 30 Hz⁽¹³⁾ and the most provocative frequencies are in the range of 15-25 Hz, although some

individuals are sensitive to single flashes or to frequencies as high as 65 Hz⁽¹⁰⁾.

JME versus AE

The cohort could be divided into three groups, depending on the type of syndrome: JME, AE and other IGE. The JME subgroup had a significant higher current age ($p=0.035$) and age at onset of seizures ($p=0.029$) compared with the AE subgroup. The AE group included all absence syndromes/types and could therefore also include patients with CAE, who in general have a younger age at onset of seizures than patients with JME.

The occurrence of sleep deprivation as a seizure trigger was reported significantly more often in the JME subgroup compared with the AE subgroup ($p=0.003$) as expected. However, sleep deprivation may precipitate seizures in all kinds of photosensitive epilepsies^(3,5,11,20).

Ictal clinical phenomena occurred significantly more in patients of the AE subgroup, compared with the JME subgroup ($p=0.016$). This could be due to a higher sensitivity of the AE patients to IPS, although no significant differences in sensitivity range, PPR type or prolonged PPR were observed between the 2 groups.

There were also no significant differences in seizure types/outcome, AED treatment status and follow-up between the AE and JME patients.

Prolonged PPR versus self-limited PPR

22 (48%) of 46 patients showed prolonged PPRs and 24 (52%) self-limited PPRs on EEG. Puglia et al⁽¹⁴⁾ concluded in their study that a prolonged PPR is associated with a higher incidence of seizures than a self-limited response, but that patients with a prolonged PPR more often had other epileptiform abnormalities than the self-limited group. In our group there were no significant differences in seizure incidence or other epileptiform abnormalities on EEG between patients with prolonged PPRs and self-limited PPRs. However, patients with prolonged PPRs experienced

significantly more clinical seizures during an EEG recording than patients with self-limited PPRs ($p=0.000$).

Factors associated with seizure outcome

The patients who were not seizure free had significantly more AS (initially) compared with the patients who became seizure free ($p=0.001$). In addition, the combination of AS+MS+GTCS (initially) occurred significantly more in this patient group ($p=0.009$) and the combination of MS+GTCS (initially) occurred significantly more in the patient group that became seizure free ($p=0.003$). These findings suggest that having AS is a negative prognostic factor for the seizure outcome. In literature, the association of photosensitivity with typical absence seizures has been shown to be a poor prognostic factor⁽⁴⁾. Similarly, Baykan et al.⁽²⁾ concluded in their study that (photosensitive) typical absence seizures are associated with an unremitting course.

This is a non-randomized retrospective cohort study with a relatively small number of patients. The AE subgroup included all types of absence syndromes (like CAE en JAE) what makes comparison and interpretation of results between subgroups difficult. So it was not clear but our findings suggest that having AS is a negative prognostic factor for the seizure outcome in patients with photosensitive IGE.

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REFERENCES

1. Anderman F, Zifkin F. The benign occipital epilepsies of childhood: an overview of the idiopathic syndromes and of the relationship to migraine. *Epilepsia*. 1998; 39, 9-23
2. Baykan B, Matur Z, Gurses C, Aykutlu E, Gokyigit A. Typical absence seizures triggered by photosensitivity. *Epilepsia*. 2005; 46, 159-163
3. Covanis A. Photosensitivity in idiopathic generalized epilepsies. *Epilepsia*. 2005; 46, 67-72
4. Covanis A, Skiadas K, Loli N, Lada C, Theodorou V. Absence epilepsy: early prognostic signs. *Seizure*. 1992; 1, 281-289
5. Fisher RS, Harding G, Erba G, Barkley GL, Wilkins A. Photic- and pattern-induced seizures: a review for the Epilepsy Foundation of America Working Group. *Epilepsia*. 2005; 46, 14261-1441
6. Guerrini R, Genton P. Epileptic syndromes and visually induced seizures. *Epilepsia*. 2004; 45, 14-18
7. Harding GF, Harding PF. Televised material and photosensitive epilepsy. *Epilepsia*. 1999; 40, 65-69
8. Harding GF, Edson A, Jeavons PM. Persistence of photosensitivity. *Epilepsia*. 1997; 38, 663-669
9. Jeavons PM, Bishop A, Harding GF. The prognosis of photosensitivity. *Epilepsia*. 1986; 27, 569-575
10. Kooi KA, Thomas MH, Mortenson FN. Photoconvulsive and photomyoclonic responses in adults. An appraisal of their clinical significance. *Neurology*. 1960; 10, 1051-1058
11. Kasteleijn-Nolst Trenite DG, Guerrini R, Binnie CD, Genton P. Visual sensitivity and epilepsy: a proposed terminology and classification for clinical and EEG phenomenology. *Epilepsia*. 2001; 42, 692-701
12. Kasteleijn-Nolst Trenite DG. Reflex seizures induced by intermittent light stimulation. *Adv Neurol*. 1998; 75, 99-121
13. Leijten FS, Dekker E, Spekreijse H, Kasteleijn-Nolst Trenite DG, Van Emde Boas W. Light diffusion in photosensitive epilepsy. *Electroencephalogr Clin Neurophysiol*. 1998; 106, 387-391
14. Puglia JF, Brenner RP, Soso MJ. Relationship between prolonged and self-limited photoparoxysmal responses and seizure incidence: study and review. *J Clin Neurophysiol*. 1992; 9, 137-144
15. Quirk JA, Fish DR, Smith SJ, Sander JW, Shorvon SD, Allen PJ. Incidence of photosensitive epilepsy: a prospective national study. *Electroencephalogr Clin Neurophysiol*. 1995; 95, 260-267
16. Rubboli G, Parra J, Seri S, Takahashi T, Thomas P. EEG diagnostic procedures and special investigations in the assessment of photosensitivity. *Epilepsia*. 2004; 45, 35-39
17. Stephani U, Tauer U, Koeleman B, Pinto D, Neubauer BA, Lindhout D. Genetics of photosensitivity (photoparoxysmal response): a review. *Epilepsia*. 2004; 45, 19-23
18. Takahashi Y, Sato T, Goto K, Fujino M, Fujiwara T, Yamaga M, et al. Optical filters inhibiting television-induced photosensitive seizures. *Neurology*. 2001; 57, 1767-1773
19. Trenite DG. Photosensitivity, visually sensitive seizures and epilepsies. *Epilepsy Res*. 2006; 70, 269-279
20. Verrotti A, Tocco AM, Salladini C, Latini G, Chiarelli F. Human photosensitivity: from pathophysiology to treatment. *Eur J Neurol*. 2005; 12, 828-841
21. Verrotti A, Basciani F, Trotta D, Cutarella R, Salladini C, Morgese G, et al. Photoparoxysmal responses in non-epileptic children in long-term follow-up. *Acta Neurol Scand*. 2002; 105, 400-402
22. Waltz S, Christen HJ, Doose H. The different patterns of the photoparoxysmal response--a genetic study. *Electroencephalogr Clin Neurophysiol*. 1992; 83, 138-145
23. Zifkin BG, Kasteleijn-Nolst Trenite D. Reflex epilepsy and reflex seizures of the visual system: a clinical review. *Epileptic Disord*. 2000; 2, 129-136